Causal Inference for the Social Sciences

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Abstract

This course introduces methods and concepts used to infer causal effects from comparisons of intervention and control groups. We'll use the potential outcomes framework of causality to show how a study's research design provides a foundation for estimation and testing. We focus, first, on properties of estimators and tests in randomized experiments, e.g., unbiasedness, consistency, controlled error rates. We then turn to research designs that are either partially controlled (e.g., experiments with noncompliance and/or attrition) or uncontrolled (e.g., observational studies). For observational studies, we focus primarily on matching methods implemented via `optmatch` and related packages in R. Finally, we turn to sensitivity analysis — namely, how to assess how inferences would change should certain assumptions about the research design be false. Examples throughout the course are drawn from economics, political science, public health, and sociology.

We assume familiarity with linear algebra and strong knowledge of statistical concepts, such as sampling distributions, statistical inference, and hypothesis testing. Demonstrations, examples and assignments make extensive use of R.

Overview

We may all warn our freshmen that association is not causation, but inferring causation has always been a central aim both for statisticians and for their collaborators. Until recently, however, inference of causation from statistical evidence depended on murky, scarcely attainable requirements; in practice, the weight of casual arguments was largely determined by the scientific authority of the people making them.

Requirements for causal inference become more clear when they are framed in terms of *potential outcomes*. This was first done by Neyman, who in the 1920s used potential outcomes to model agricultural experiments. Fisher independently proposed a related but distinct, ultimately more influential, analysis of experiments in 1935, and a rich strain of causal analysis developed among his intellectual progeny. It clarified the differing requirements for causal inference with experiments and with observational data, isolating the distinct contributions required of the statistician and of his disciplinary collaborators; generated more satisfying methods with which to address potential confounding due to measured variables; qualitatively and quantitatively advanced our grasp of unmeasured confounding and its potential ramifications; furnished statistical methods with which to eke more out of the strongest study designs, under fewer assumptions; and articulated principles with which to understand study designs as a spectrum, rather than a dichotomy between "good" experiments and "bad" observational studies. Understanding the methods and outlook of the school founded by Fisher's student W. G. Cochran will be the central task of this course.

The course begins by applying the Fisher and Neyman-Rubin approaches to statistical inference for counterfactual causal effects to randomized experiments, touching on considerations specific to clustered treatment assignment, "small" sample sizes and treatment effect heterogeneity. The next segment addresses conceptual and methodological challenges of applying the same models to analysis of non-experimental data. This course segment covers ignorability, selection, "common support," covariate balance, paired comparisons, optimal matching and propensity scores. With these foundations in place, the course then turns to sensitivity analysis — namely, how to make principled assessments of how inferences would change should crucial assumptions be false. Over the course's three weeks, the course becomes progressively less conceptual and more applied with increasing emphasis on computing strategies in R.

Administrative

Textbooks

The main texts for the course are

Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press

Paul R. Rosenbaum (2010). *Design of Observational Studies.* New York, NY: Springer

Paul R. Rosenbaum (2002b). *Observational Studies.* Second. New York, NY: Springer

These three textbooks are presented in varying difficulty and we will draw from all three. Although we won't follow these books closely, their goals and methods align with the course's, and they will be useful as references and supplements.

Other texts that we draw on include

Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton

Guido W. Imbens and Donald B. Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction.* New York, NY: Cambridge University Press

Peng Ding (2024). *A First Course in Causal Inference.* Chapman and Hall/CRC

Paul R. Rosenbaum (2025). *An Introduction to the Theory of Observational Studies.* Springer Texts in Statistics. New York, NY: Springer

Other readings will be assigned and distributed electronically.

If you're new to R, we suggest getting a hold of:

John Fox (2016). *Applied Regression Analysis and Generalized Linear Models.* 3rd. Los Angeles, CA: SAGE Publications

Hadley Wickham and Garrett Grolemund (2017). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data.* First. Sebastopol, CA: O'Reilly Media

R software will be required for several specific segments of the course. With some independent effort, students not familiar with R in advance should be able to learn enough R during the course to complete these assignments. We also recommend some work with R — for example, via working through some online R courses — before the course for students who have never used it before.

Assignments

Assignments are due each Friday, at the beginning of class. Parts of the assignment will be given at the beginning of the week, but other parts will be given during class, over the course of the week. Late homework will not be accepted without cause (or prior arrangement with a teaching assistant).

Participation is expected. It can take various forms:

1. Doing in-class exercises and discussing them with your peers;
2. From time to time, making a clarification or raising a clarifying question;
3. Contributing to in-class discussions;
4. Drop by one of the professor's office hours to share a point that you *and at least one classmate* would like to have clarified or amplified, or to point out a connection to your field;

If you are taking the course for a grade, make a point of doing at least one of 3 and 4.

Course Schedule

The schedule appears below; the Course content section gives fuller readings for each topic, plus the menu of student-chosen special topics for the final day.

Table 1: Course schedule by day, instructor, topic, readings, and application

Day	Instructor	Topic	Required readings	Application
Day 1	Tom	Introduction: Random allocation, potential outcomes, and Fisher's exact test	Holland (1986, § 1–4) Rosenbaum (2017, Chapter 2)	Fisher (1935)
Day 2	Tom	The sharp framework: Fisherian inference	Rosenbaum (2017, Chapter 3) Fisher (1935, § 5–10)	Arceneaux (2005)
Day 3	Tom	Weak framework I: Estimating average effects and the variance of the difference-in-means	Gerber and Green (2012, Ch. 2 and pp. 51–61) Aronow and Middleton (2013) Middleton and Aronow (2015) Freedman et al. (1998) Imbens and Rubin (2015, pp. 87–98)	
Day 4	Tom	Weak framework II: Conservative variance estimation, hypothesis testing, and covariance adjustment	Gerber and Green (2012, Ch. 3–4) Rosenbaum (2002); Lin (2013)	
Day 5 (HW 1 due)	–	No Class: U.S. Holiday		
Day 6	Tom	Noncompliance and attrition	Gerber and Green (2012, Ch. 5–7) Rosenbaum (1996) Rosenbaum (2010, § 5.3)	Albertson and Lawrence (2009)
Day 7	Tom	Observational studies and as-if randomization	Leavitt and Miratrix (2026, § 1) Rosenbaum (2017, Chapter 5) Gelman and Hill (2006, § 9.0–9.2) Berk (2010)	Gilligan and Sergenti (2008); Cerdá et al. (2012)
Day 8	Tom	Building a matched design: Distances and forming matches	Leavitt and Miratrix (2026, § 2.1–2.2.4) Rosenbaum (2017, pp. 65–96) Rosenbaum (2010, Ch. 7–8) Hansen (2011)	
Day 9	Tom	Evaluating a matched design: Balance and effective sample size	Leavitt and Miratrix (2026, § 2.2.5) Hansen and Bowers (2008)	
Day 10 (HW 2 due)	Tom	Inference under as-if randomization I: The sharp framework	Leavitt and Miratrix (2026, § 2.3–2.3.1) Rosenbaum (2017, Chapter 3)	
Day 11	Jake	Inference under as-if randomization II: The weak framework	Leavitt and Miratrix (2026, § 2.3.2) Gerber and Green (2012, pp. 71–79) Fogarty (2018); Pashley and Miratrix (2020)	
Day 12	Jake	Sensitivity analysis I: The sharp framework	Leavitt and Miratrix (2026, § 2.4–2.4.1) Rosenbaum (2017, Chapter 9) Rosenbaum (2018)	
Day 13	Jake	Sensitivity analysis II: The weak framework	Leavitt and Miratrix (2026, § 2.4.2) Fogarty (2023)	
Day 14 (HW 3 due)	Jake	Full pipeline end-to-end and/or special topics (students' choice)	Leavitt and Miratrix (2026)	

Course content

1 Randomized experiments

1.1 Introduction: Random allocation, potential outcomes and Fisher's exact test

Required

Paul W. Holland (1986). "Statistics and Causal Inference". In: *Journal of the American Statistical Association* 81.396, pp. 945–960, Sections 1 – 4.

Chapter 2 of Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press.

Recommended

Section 1.2, "Experimentation defined," of Donald R. Kinder and Thomas R. Palfrey (1993). "On Behalf of an Experimental Political Science". In: *Experimental Foundations of Political Science.* Ed. by Donald R. Kinder and Thomas R. Palfrey. Michigan Studies in Political Analysis. Ann Arbor, MI: University of Michigan Press. Chap. 1, pp. 1–39. (Particularly pp. 5 – 10.)

Chapter 1, "Introduction," of Ronald Aylmer Fisher (1935). *The Design of Experiments.* Edinburgh, SCT: Oliver and Boyd and pages 131 – 135 of Joan Fisher Box (1978). *R. A. Fisher, the Life of a Scientist.* New York, NY: Wiley for historical context.

Jersey Neyman (1923). "Sur les applications de la théorie des probabilités aux experiences agricoles: Essai des principes". In: *Roczniki Nauk Rolniczych* 10, pp. 1–51

Donald B. Rubin (1974). "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies". In: *Journal of Educational Psychology* 66.5, p. 688

Jake Bowers and Thomas Leavitt (2020). "Causality and Design-Based Inference". In: *The SAGE Handbook of Research Methods in Political Science and International Relations.* Ed. by Luigi Curini and Robert Franzese. Vol. 2. Thousand Oaks, CA: SAGE Publications. Chap. 41, pp. 769–804

1.2 The sharp framework: Fisherian inference for causal effects

Application

Kevin Arceneaux (2005). "Using Cluster Randomized Field Experiments to Study Voting Behavior". In: *The Annals of the American Academy of Political and Social Science* 601.1, pp. 169–179

Required

Chapter 3 of Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press.

Sections 5 – 10 (pp. 11 – 19) of Ronald Aylmer Fisher (1935). *The Design of Experiments.* Edinburgh, SCT: Oliver and Boyd.

Recommended

Pages 27 – 49 of Paul R. Rosenbaum (2002b). *Observational Studies.* Second. New York, NY: Springer

Chapter 2 of Paul R. Rosenbaum (2010). *Design of Observational Studies.* New York, NY: Springer.

Devin Caughey, Allan Dafoe, Xinran Li, and Luke W. Miratrix (2023). "Randomisation Inference Beyond the Sharp Null: Bounded Null Hypotheses and Quantiles of Individual Treatment Effects". In: *Journal of the Royal Statistical Society Series B (Statistical Methodology)* 85.5, pp. 1471–1491

David Kim, Yongchang Su, Jake Bowers, and Xinran Li (2026). "Randomization Tests for Distributions of Individual Treatment Effects via Combined Rank Statistics". ArXiv preprint, arXiv:2605.08027

Andrew Gelman (2003). "A Bayesian Formulation of Exploratory Data Analysis and Goodness-of-fit Testing". In: *International Statistical Review* 71.2, pp. 369–382

1.3 The weak framework I: Estimating average effects and the variance of the difference-in-means

Required

Chapter 2 and pp. 51 – 61 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton

Peter M. Aronow and Joel A. Middleton (2013). "A Class of Unbiased Estimators of the Average Treatment Effect in Randomized Experiments". In: *Journal of Causal Inference* 1.1, pp. 135–154

Joel A. Middleton and Peter M. Aronow (2015). "Unbiased Estimation of the Average Treatment Effect in Cluster-Randomized Experiments". In: *Statistics, Politics and Policy* 6.1-2, pp. 39–75

Endnote spanning pages A-32 and 33, David A. Freedman, Robert Pisani, and Roger Purves (1998). *Statistics.* 3rd. New York, NY: W. W. Norton & Company. (This can be read as a précis of: Jersey Neyman (1923). "Sur les applications de la théorie des probabilités aux experiences agricoles: Essai des principes". In: *Roczniki Nauk Rolniczych* 10, pp. 1–51.)

Chapter 6, pp. 87 – 98 of Guido W. Imbens and Donald B. Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction.* New York, NY: Cambridge University Press

1.4 The weak framework II: Conservative variance estimation and hypothesis testing

Required

Chapter 3 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton.

Recommended

Chapter 6 of Thad Dunning (2012). *Natural Experiments in the Social Sciences: A Design-Based Approach.* New York, NY: Cambridge University Press

Xinran Li and Peng Ding (2017). "General Forms of Finite Population Central Limit Theorems with Applications to Causal Inference". In: *Journal of the American Statistical Association* 112.520, pp. 1759–1769

Peng Ding (2017a). "A Paradox from Randomization-Based Causal Inference". In: *Statistical Science* 32.3, pp. 331–345

Peter M. Aronow, Donald P. Green, and Donald K. K. Lee (2014). "Sharp Bounds on the Variance in Randomized Experiments". In: *The Annals of Statistics* 42.3, pp. 850–871

James M. Robins (1988). "Confidence Intervals for Causal Parameters". In: *Statistics in Medicine* 7.7, pp. 773–785

Christopher Harshaw, Joel A. Middleton, and Fredrik Sävje (2026). "Optimized Variance Estimation under Interference and Complex Experimental Designs". In: *Journal of the American Statistical Association*

1.5 Covariance adjustment (sharp and weak frameworks)

Required

Chapter 4 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton.

Paul R. Rosenbaum (2002a). "Covariance Adjustment in Randomized Experiments and Observational Studies". In: *Statistical Science* 17.3, pp. 286–327.

Winston Lin (2013). "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique". In: *The Annals of Applied Statistics* 7.1, pp. 295–318

Recommended

Luke W. Miratrix, Jasjeet S. Sekhon, and Bin Yu (2013). “Adjusting Treatment Effect Estimates by Post-Stratification in Randomized Experiments”. In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 75.2, pp. 369–396

David A. Freedman (2008b). “On Regression Adjustments to Experimental Data”. In: *Advances in Applied Mathematics* 40.2, pp. 180–193

David A. Freedman (2008c). “Randomization Does Not Justify Logistic Regression”. In: *Statistical Science* 23.2, pp. 237–249

David A. Freedman (2008a). “On Regression Adjustments in Experiments with Several Treatments”. In: *The Annals of Applied Statistics* 2.1, pp. 176–196

Cyrus Samii and Peter M. Aronow (2012). “On Equivalencies between Design-based and Regression-based Variance Estimators for Randomized Experiments”. In: *Statistics & Probability Letters* 82.2, pp. 365–370

Peter M. Aronow and Cyrus Samii (2016). “Does Regression Produce Representative Estimates of Causal Effects?” In: *American Journal of Political Science* 60.1, pp. 250–267

Alberto Abadie, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge (2020). “Sampling-Based versus Design-Based Uncertainty in Regression Analysis”. In: *Econometrica* 88.1, pp. 265–296

Colin B. Fogarty (2018b). “Regression-assisted Inference for the Average Treatment Effect in Paired Experiments”. In: *Biometrika* 105.4, pp. 994–1000

Kevin Guo and Guillaume W. Basse (2023). “The Generalized Oaxaca-Blinder Estimator”. In: *Journal of the American Statistical Association* 118.541, pp. 524–536

Peter L. Cohen and Colin B. Fogarty (2023). “No-Harm Calibration for Generalized Oaxaca-Blinder Estimators”. In: *Biometrika* 111.1, pp. 331–338

Haoge Chang, Joel A. Middleton, and P. M. Aronow (2024). “Exact Bias Correction for Linear Adjustment of Randomized Controlled Trials”. In: *Econometrica* 92, pp. 1503–1519

1.6 Noncompliance and attrition

Application

Bethany Albertson and Adria Lawrence (2009). “After the Credits Roll: The Long-Term Effects of Educational Television on Public Knowledge and Attitudes”. In: *American Politics Research* 37.2, pp. 275–300

1.6.1 Noncompliance and instrumental variables

Required

Chapters 5 and 6 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton.

Section 5.3, “Instruments,” of Paul R. Rosenbaum (2010). *Design of Observational Studies.* New York, NY: Springer

Paul R. Rosenbaum (1996). “Identification of Causal Effects Using Instrumental Variables: Comment”. In: *Journal of the American Statistical Association* 91.434, pp. 465–468

Recommended

Section 2.3 of Paul R. Rosenbaum (2002a). “Covariance Adjustment in Randomized Experiments and Observational Studies”. In: *Statistical Science* 17.3, pp. 286–327

Joshua D. Angrist, Guido W. Imbens, and Donald B. Rubin (1996). “Identification of Causal Effects Using Instrumental Variables”. In: *Journal of the American Statistical Association* 91.434, pp. 444–455

Guido W. Imbens and Paul R. Rosenbaum (2005). “Robust, Accurate Confidence Intervals with a Weak Instrument: Quarter of Birth and Education”. In: *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 168.1, pp. 109–126

Hyunseung Kang, Laura Peck, and Luke Keele (2018). “Inference for Instrumental Variables: A Randomization Inference Approach”. In: *Journal of the Royal Statistical Society. Series A: Statistics in Society* 181.4, pp. 1231–1254

Ben B. Hansen and Jake Bowers (2008). “Covariate Balance in Simple, Stratified and Clustered Comparative Studies”. In: *Statistical Science* 23.2, pp. 219–236

P. M. Aronow, Haoge Chang, and Patrick Lopatto (2025). “Randomization-Based Confidence Sets for the Local Average Treatment Effect”. In: *Biometrika.* In press

Nicole E. Pashley (2022). “Note on the Delta Method for Finite Population Inference with Applications to Causal Inference”. In: *Statistics & Probability Letters* 188, p. 109540

Nicole E. Pashley, Luke Keele, and Luke W. Miratrix (2024). “Improving Instrumental Variable Estimators with Poststratification”. In: *Journal of the Royal Statistical Society Series A: Statistics in Society* 188.3, pp. 765–790

Matthew Blackwell and Nicole E. Pashley (2021). “Noncompliance and Instrumental Variables for 2^K Factorial Experiments”. In: *Journal of the American Statistical Association* 118.542, pp. 1102–1114

1.6.2 Attrition, or missing outcomes

Required

Chapter 7 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton

Recommended

David S. Lee (2009). "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects". In: *The Review of Economic Studies* 76.3, pp. 1071–1102

Peter M. Aronow, Jonathon Baron, and Lauren Pinson (2019). "A Note on Dropping Experimental Subjects who Fail a Manipulation Check". In: *Political Analysis* 27.4, pp. 572–589

Joel L. Horowitz and Charles F. Manski (2000). "Nonparametric Analysis of Randomized Experiments with Missing Covariate and Outcome Data". In: *Journal of the American Statistical Association* 95.449, pp. 77–84

Alexander Coppock, Alan S. Gerber, Donald P. Green, and Holger L. Kern (2017). "Combining Double Sampling and Bounds to Address Nonignorable Missing Outcomes in Randomized Experiments". In: *Political Analysis* 25.2, pp. 188–206

Xinran Li, Peizan Sheng, and Zeyang Yu (2025). "Randomization Inference with Sample Attrition". ArXiv preprint, arXiv:2507.00795

Haoge Chang and Zeyang Yu (2026). "Randomization Inference For the Always-Reporter Average Treatment Effect". ArXiv preprint, arXiv:2603.24970

2 Observational studies

2.1 Observational studies and as-if randomization

Required

Section 1, "Design-Based Foundations of the Matching Pipeline," of Thomas Leavitt and Luke W. Miratrix (2026). "Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R". Accepted at *Observational Studies*

Chapter 5 of Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press

Sections 9.0 – 9.2 (especially discussion of interpolation and extrapolation) of Andrew Gelman and Jennifer Hill (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models.* New York, NY: Cambridge University Press

Richard A. Berk (2010). "What You Can and Can't Properly Do with Regression". In: *Journal of Quantitative Criminology* 26.4, pp. 481–487

Recommended

Marie-Abele C. Bind and Donald B. Rubin (2019). "Bridging Observational Studies and Randomized Experiments by Embedding the Former in the Latter". In: *Statistical Methods in Medical Research* 28.7, pp. 1958–1978

Donald B. Rubin (1977). "Assignment to Treatment Group on the Basis of a Covariate". In: *Journal of Educational Statistics* 2.1, pp. 1–26

William G. Cochran (1965). "The Planning of Observational Studies of Human Populations". In: *Journal of the Royal Statistical Society. Series A (General)* 128.2, pp. 234–266

Christopher H. Achen (2002). "Toward a New Political Methodology: Microfoundations and ART". in: *Annual Review of Political Science* 5, pp. 423–450

Chapters 11 and 19 (on overly influential points) of John Fox (2016). *Applied Regression Analysis and Generalized Linear Models.* 3rd. Los Angeles, CA: SAGE Publications

2.2 Building a matched design: Measuring similarity and forming matches

Application

Michael J. Gilligan and Ernest J. Sergenti (2008). "Do UN Interventions Cause Peace? Using Matching to Improve Causal Inference". In: *Quarterly Journal of Political Science* 3.2, pp. 89–122

Magdalena Cerdá, Jeffrey D. Morenoff, Ben B. Hansen, Kimberly J. Tessari Hicks, Luis F. Duque, Alexandra Restrepo, and Ana V. Diez-Roux (2012). "Reducing Violence by Transforming Neighborhoods: A Natural Experiment in Medellín, Colombia". In: *American Journal of Epidemiology* 175.10, pp. 1045–1053

Required

Sections 2.1 – 2.2.4 (running example through forming the matches) of Thomas Leavitt and Luke W. Miratrix (2026). "Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R". Accepted at *Observational Studies*

Pages 65 – 96 of Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press

Chapters 7 – 8 of Paul R. Rosenbaum (2010). *Design of Observational Studies.* New York, NY: Springer

Ben B. Hansen (2011). "Propensity Score Matching to Extract Latent Experiments from Nonexperimental Data: A Case Study". In: *Looking Back: Proceedings of a Conference in Honor of Paul W. Holland.* Ed. by Neil J. Dorans and Sandip Sinharay. Vol. 202. Lecture Notes in Statistics. New York, NY: Springer. Chap. 9, pp. 149–181

Recommended

Ben B. Hansen (2004). "Full Matching in an Observational Study of Coaching for the SAT". in: *Journal of the American Statistical Association* 99.467, pp. 609–618

Ben B. Hansen and Stephanie Olsen Klopfer (2006). "Optimal Full Matching and Related Designs via Network Flows". In: *Journal of Computational and Graphical Statistics* 15.3, pp. 609–627

Xing Sam Gu and Paul R. Rosenbaum (1993). "Comparison of Multivariate Matching Methods: Structures, Distances, and Algorithms". In: *Journal of Computational and Graphical Statistics* 2.4, pp. 405–420

Donald B. Rubin and Richard P. Waterman (2006). "Estimating the Causal Effects of Marketing Interventions Using Propensity Score Methodology". In: *Statistical Science* 21.2, pp. 206–222

Ben B. Hansen (2008b). "The Prognostic Analogue of the Propensity Score". In: *Biometrika* 95.2, pp. 481–488

Adam C. Sales, Ben B. Hansen, and Brian Rowan (2018). "Rebar: Reinforcing a Matching Estimator With Predictions From High-Dimensional Covariates". In: *Journal of Educational and Behavioral Statistics* 43.1, pp. 3–31

Paul R. Rosenbaum (2020). "Modern Algorithms for Matching in Observational Studies". In: *Annual Review of Statistics and Its Application* 7.1, pp. 143–176

Chapter 3 of Paul R. Rosenbaum (2002b). *Observational Studies.* Second. New York, NY: Springer, specifically Sections 3.1 – 3.2 and 3.4 – 3.5.

Paul R. Rosenbaum (2001b). "Observational Studies: Overview". In: *International Encyclopedia of the Social & Behavioral Sciences.* Ed. by Neil J. Smelser and Paul B. Baltes. Elsevier/North-Holland [Elsevier Science Publishing Co., New York; North-Holland Publishing Co., Amsterdam], pp. 10808–10815

Robert Bifulco (2012). "Can Nonexperimental Estimates Replicate Estimates Based on Random Assignment in Evaluations of School Choice? A Within-Study Comparison". In: *Journal of Policy Analysis and Management* 31.3, pp. 729–751

Kevin Arceneaux (2010). "A Cautionary Note on the Use of Matching to Estimate Causal Effects: An Empirical Example Comparing Matching Estimates to an Experimental Benchmark". In: *Sociological Methods & Research* 39.2, pp. 256–282

Donald B. Rubin (1979). "Using Multivariate Matched Sampling and Regression Adjustment to Control Bias in Observational Studies". In: *Journal of the American Statistical Association* 74.366a, pp. 318–328

James M. Robins, Miguel Ángel Hernán, and Babette Brumback (2000). "Marginal Structural Models and Causal Inference in Epidemiology". In: *Epidemiology* 11.5, pp. 550–560

Daniel E. Ho, Kosuke Imai, Gary King, and Elizabeth A. Stuart (2007). “Matching as Nonparametric Preprocessing for Reducing Model Dependence in Parametric Causal Inference”. In: *Political Analysis* 15.3, pp. 199–236

Chapter 13 of Paul R. Rosenbaum (2010). *Design of Observational Studies.* New York, NY: Springer

Paul R. Rosenbaum and Donald B. Rubin (1985). “Constructing a Control Group Using Multivariate Matched Sampling Methods that Incorporate the Propensity Score”. In: *The American Statistician* 39.1, pp. 33–38

2.3 Evaluating a matched design: Balance and effective sample size

Required

Section 2.2.5 (balance and effective sample size) of Thomas Leavitt and Luke W. Miratrix (2026). “Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R”. Accepted at *Observational Studies*

Ben B. Hansen and Jake Bowers (2008). “Covariate Balance in Simple, Stratified and Clustered Comparative Studies”. In: *Statistical Science* 23.2, pp. 219–236

Recommended

Peter C. Austin (2009). “Balance Diagnostics for Comapring the Distribution of Baseline Covariates between Treatment Groups in Propensity-Score Matched Samples”. In: *Statistics in Medicine* 28.25, pp. 3083–3107

Elizabeth A. Stuart (2010). “Matching Methods for Causal Inference: A Review and a Look Forward”. In: *Statistical Science* 25.1, p. 1

Donald B. Rubin (2001). “Using Propensity Scores to Help Design Observational Studies: Application to the Tobacco Litigation”. In: *Health Services and Outcomes Research Methodology* 2.3, pp. 169–188

Peter C. Austin (2008). “A Critical Appraisal of Propensity-Score Matching in the Medical Literature between 1996 and 2003”. In: *Statistics in Medicine* 27.12, pp. 2037–2049

Ben B. Hansen (2008a). “The Essential Role of Balance Tests in Propensity-Matched Observational Studies: Comments on ‘A Critical Appraisal of Propensity-Score Matching in the Medical Literature between 1996 and 2003’ by Peter Austin”. In: *Statistics in Medicine* 27.12, pp. 2050–2054

Sture Holm (1979). “A Simple Sequentially Rejective Multiple Test Procedure”. In: *Scandinavian Journal of Statistics* 6.2, pp. 65–70

Kosuke Imai (2008). “Variance Identification and Efficiency Analysis in Randomized Experiments under the Matched-Pair Design”. In: *Statistics in Medicine* 27.24, pp. 4857–4873

Kosuke Imai, Gary King, and Elizabeth A. Stuart (2008). “Misunderstandings between Experimentalists and Observationalists about Causal Inference”. In: *Journal of* *the Royal Statistical Society: Series A (Statistics in Society)* 171.2, pp. 481–502

2.4 Inference under as-if randomization I: The sharp framework

Required

Sections 2.3 – 2.3.1 (inference under the sharp framework) of Thomas Leavitt and Luke W. Miratrix (2026). "Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R". Accepted at *Observational Studies*

Chapter 3 of Paul R. Rosenbaum (2017). *Observation and Experiment: An Introduction to Causal Inference.* Cambridge, MA: Harvard University Press

Recommended

Nicole E. Pashley, Guillaume W. Basse, and Luke W. Miratrix (2021). "Conditional as-if analyses in randomized experiments". In: *Journal of Causal Inference* 9.1

2.5 Inference under as-if randomization II: The weak framework

Required

Section 2.3.2 (inference under the weak framework) of Thomas Leavitt and Luke W. Miratrix (2026). "Building a Design-Based Matching Pipeline: From Principles to Practical Implementation in R". Accepted at *Observational Studies*

Pages 71 – 79 of Alan S. Gerber and Donald P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation.* New York, NY: W.W. Norton

Colin B. Fogarty (2018a). "On Mitigating the Analytical Limitations of Finely Stratified Experiments". In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 80.5, pp. 1035–1056

Nicole E. Pashley and Luke W. Miratrix (2020). "Insights on Variance Estimation for Blocked and Matched Pairs Designs". In: *Journal of Educational and Behavioral Statistics*

Recommended

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